# 3-PAGE PROJECT PROPOSAL

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Abstract. This is a 3-page project proposal for the fulfillment of part of the requirements to reach candidacy. We identify an unsolved research problem, list some of our results addressing that problem, and briefly discuss our future goals towards the problem and related areas. This project is being advised by Egon Schulte.

## 1. Preliminary notions

We observe the conventions and results of [1–3]

1.1. **Abstract polytopes.** An *abstract $n$-polytope* (or just *polytope*) is a ranked poset satisfying four properties arising naturally from a geometric setting, and an *automorphism* of a polytope $\mathcal{P}$ is an incidence-preserving bijection $\mathcal{P} \longrightarrow \mathcal{P}$. The collection of automorphisms of $\mathcal{P}$ form a group $\Gamma(\mathcal{P})$, and this group acts freely on the *flags* of $\mathcal{P}$. We say $\mathcal{P}$ is *regular* when the action of $\Gamma(\mathcal{P})$ on the flags of $\mathcal{P}$ is also transitive. When the action of $\Gamma(\mathcal{P})$ induces exactly two orbits on the flags of $\mathcal{P}$ in such a way that flags differing in exactly one element lie in distinct orbits, we say $\mathcal{P}$ is *chiral*. It is well-known that the groups of regular polytopes correspond to *string C-groups*. In fact, the groups of chiral polytopes correspond to *string even groups*, of which the even subgroups of string C-groups are an example.

1.2. **Basal polytopes.** A polytope $\mathcal{P}$ is said to *cover* a polytope $\mathcal{Q}$ if $\mathcal{Q}$ is the orbit space $\mathcal{P}/N$ for some subgroup $N \leq \Gamma(\mathcal{P})$. Both regular and chiral polytopes are covered by a universal polytope $\mathcal{U}$ of a given type. In fact, a polytope $\mathcal{U}/N$ is regular if and only if $N$ is normal in $\Gamma(\mathcal{U})$. On the other hand, a polytope $\mathcal{U}/N$ is chiral if and only if it is not regular and $N$ is normal in $\Gamma^+(\mathcal{U})$.

Of interest to us in this research project are those polytopes that are minimal in the covering lattice, which we've named the class of *basal* polytopes. Other classes of minimal polytopes have been explored. The so-called *tight* polytopes are those of a fixed type with the minimal number of flags, and there have been many papers published on them (see [4–7] for a few examples). To our knowledge, however, nothing has been written on the basal polytopes.

## 2. Proposed research topic

In his atlas of regular polytopes of order at most 2000 (see [8]), Hartley notes that enumerating groups of order 1536 (of which there are over 400 million) and 1024 (of which there are over 49 billion) requires developing a theory characterizing

string C-groups of order $2^k$ and $2^k p$ for prime $p$. This call to action is one of the open problems listed in [9], and is what we hope to address in our project.

**Problem 1** (Problem 30 in [9])**.** Characterize the groups of regular or chiral polytopes of order $2^k$ or $2^k p$ where $k$ is a positive integer and $p$ an odd prime.

There have been many papers addressing aspects of problem 1 for the regular case. See [10–13] for constructions of regular polytopes of order $2^k$. To our knowledge, the only paper addressing problem 1 for chiral polytopes is [14]. We use the RAMP package [15] in GAP [16] to gather evidence and make conjectures before proving theorems.

## 3. Progress

To address problem 1, we introduce the notion of a $\lambda$-*group*, which prescribes to finitely-generated subgroups a *winding period*. Let $\Gamma$ be a group with $A \leq \Gamma$ a subgroup generated by elements $A_1, \ldots, A_n$. For $p$ an odd prime and unit $j \in (\mathbb{Z}/p\mathbb{Z})^\times$, the $\lambda$-*group* of $A$ with *winding period* $p$ is the subgroup $\Lambda(A,p)$ generated by elements $\prod_{i=1}^n A_i^{\ell_i}$ in $A$ for which the identity $\sum_{i=1}^n \ell_i j^{i-1} \equiv 0 \bmod p$ holds.

3.1. **Result for $\lambda$-groups.** When $\Gamma$ is the group of the universal map $\{4\alpha, 4\alpha\}$ for some $\alpha \geq 1$, we take $A = \langle A_1, \ldots, A_{4\alpha}\rangle$ to be the *translation subgroup* of $\Gamma$ generated by *glide translations*; see [17]. These glide translations obey certain fundamental relations with respect to the rotations generating the subgroup $\Gamma^+$. In fact, these relations were the motivation behind the $\lambda$-group construction presented below.

**Proposition 1.** *Let $\alpha \geq 1$. For an odd prime $p$ and unit $j \in (\mathbb{Z}/p\mathbb{Z})^\times$ satisfying*

$$j^{2\alpha} \equiv -1 \bmod p,$$

*the $\lambda$-group $\Lambda(A,p)$ is normal in $[4\alpha, 4\alpha]^+$, but fails to be so in $[4\alpha, 4\alpha]$.*

The following corollary is immediate.

**Corollary 1.** *The group $[4\alpha, 4\alpha]^+/\Lambda(A,p)$ is isomorphic to the group of a chiral polyhedron.*

3.2. **Problem 1 for basal chiral polyhedra.** Currently, we have three primary results for chiral polyhedra of order $2^k p$: two for those which are basal, and one for those of type $\{2^{k_1}, 2^{k_2}\}$ for $k_1, k_2 \geq 2$.

**Theorem 1.** *For $\mathcal{P}$ a basal chiral polyhedron of order $2^k p$, the following are equivalent:*

*(i)* $p \equiv 1 \bmod 2^k$.
*(ii)* $\mathcal{P}$ *is of type* $\{2^k, -\}$ *or* $\{-, 2^k\}$.
*(iii)* $\mathcal{P}$ *is of type* $\{2^k, 2^k\}$.

Paradoxically, a search in GAP reveals more than half of all basal chiral polyhedra of order $2^k p \leq 2000$ are those for which item (iii) holds despite item (i) seeming rare for primes $p$ as $k$ grows. We used theorem 1, a result on the necessary structure of a group of order $2^k p$, as well as a $\lambda$-group construction to fully classify the basal chiral polyhedra of order $2^k p$ with $p \equiv 1 \bmod 2^k$.

**Theorem 2.** *For every $k \geq 2$ and every odd prime $p \equiv 1 \bmod 2^k$, there exists exactly $2^{k-2}$ distinct basal chiral polyhedra of order $2^k p$, each with group $C_p \rtimes C_{2^k}$. Theorem 1 shows that these are the only basal chiral polyhedra of type $\{2^k, 2^k\}$.*

3.3. **Problem 1 for chiral polyhedra of type $\{2^{k_1}, 2^{k_2}\}$.** Our final primary theorem concerns this class of chiral polyhedra of order $2^k p$.

**Theorem 3.** *Fix $k_1, k_2 \geq 2$. Every chiral polyhedron of order $2^k p$ and type $\{2^{k_1}, 2^{k_2}\}$ covers a unique basal chiral polyhedron, namely, one of those in theorem 2.*

Constructing new chiral polyhedra of order $2^n p$ for which $p$ divides neither entry in the Schläfli type then reduces to a subset of the general group extension problem since these new polyhedra have groups which are isomorphic to extensions of $C_p \rtimes C_{2^k}$ by 2-groups. We are interested in characterizing these extensions further, as it appears relatively simple to satisfy the intersection condition necessary for chirality. For example, it turns out that trivial extensions of this semidirect product by cyclic 2-groups, *i.e.*, those groups of the form

$$(C_p \rtimes C_{2^k}) \times C_{2^n} \cong C_p \rtimes (C_{2^k} \times C_{2^n}),$$

are the groups of chiral polyhedra. The same is true of trivial extensions by dihedral groups which need not necessarily be 2-groups.

3.4. **Connection to affine groups.** A theorem of Burnside [18] shows that a finite transitive permutation group of prime degree $p$ is either doubly-transitive or isomorphic to a subgroup of the general affine group $\mathrm{AGL}(1, p) \cong \mathbb{F}_p \rtimes \mathbb{F}_p^\times$. Using theorem 1, it is simple to see that basal chiral polyhedra of order $2^k p$ satisfy the conditions of Burnside's theorem when $p \equiv 1 \bmod 2^k$, and we've shown that they are not doubly-transitive. This offers another perspective to the construction of theorem 2 beyond the use of $\lambda$-groups. It also suggests that there is a high density of chiral polyhedra which are extended from the groups of basal chiral polyhedra corresponding to Fermat primes, which is supported by evidence in GAP.

We're interested in developing these ideas further, as well as understanding which subgroups of $\mathrm{AGL}(1, p^n)$ are isomorphic to the groups of chiral polyhedra. We also believe there may be some aspect of these affine groups which may give insight into the polytope extension problem. We are also currently writing a paper [19] on this topic, and our investigation is leading us to search for $(d+2)$-polytopes whose groups are $\mathrm{AGL}(d, p^n)$, be they chiral, regular, or neither.

## References


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